# A Tunable GP-HOCBF Safety Filter for Supercritical Boiler-Turbine Control under Model Mismatch

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**Abstract**

Ultra-supercritical boiler-turbine units must keep main steam pressure, separator outlet enthalpy, and turbine power inside protection limits under coal-quality drift, fouling, actuator degradation, and other model mismatch. RoCBF-Net is a tunable safety-filter architecture for a 1000 MW boiler-turbine measurement and control loop. At each sampling instant, an upstream controller proposes an action; a Gaussian-process (GP)-corrected high-order control barrier function (HOCBF) layer then solves a quadratic program to return the nearest action satisfying the process safety constraints. The implementation propagates GP posterior uncertainty through the HOCBF $\psi$-chain and scales the resulting compositional margin with a scalar factor $\epsilon_\kappa$, so the robustness margin can be commissioned rather than fixed a priori. Evaluation uses a 5th-order nonlinear boiler-turbine benchmark with $\Phi$-scaled control effectiveness, six disturbance scenarios, load-following operation, eight controller/filter variants, and 5 seeds $\times$ 50 episodes $\times$ 500 steps. GP mean correction reduces nominal HOCBF violations by more than 500 times and achieves $\leq 0.18\%$ violation in six of seven uncertainty scenarios. Margin calibration is condition-dependent: $\epsilon_\kappa = 0$ is sufficient for additive perturbations, $\epsilon_\kappa = 0.1$ restores zero violation under state-dependent coupling, and the full theoretical margin $\epsilon_\kappa = 1.0$ over-constrains the quadratic program in the tested benchmark. A coupling-strength sweep defines a deployment envelope in which positive margins help at moderate coupling but should be disabled beyond $\gamma \geq 1.5$. The JIT-compiled filter runs at about 25 ms per step, within the 1 s sampling budget and approximately ten times faster than the implemented SLSQP-based NMPC benchmark. The results position RoCBF-Net as a pre-deployment, commissioning-calibrated safety layer for industrial process-control systems, not as a fixed worst-case robust controller.

**Keywords:** Supercritical boiler-turbine control; safety filter; high-order control barrier function; Gaussian process; model mismatch; robustness margin

# 1 Introduction

## 1.1 Industrial Measurement and Control Context

Ultra-supercritical thermal power units operate close to equipment protection boundaries. In a 1000 MW boiler-turbine unit, the control system continuously measures and regulates main steam pressure $p_m$, separator outlet enthalpy $h_m$, turbine electrical power $N_e$, boiler gas-side state $r_B$, and fuel-delay state $\tau_f$. The practical limits considered here are pressure 13–24 MPa, separator enthalpy 2670–2830 $\mathrm{kJ\,kg^{-1}}$, and power output within $\pm 50$ MW of the operating target. These limits correspond to protection, tracking, actuator, and coordinated-control requirements that a distributed control system (DCS) must satisfy while following the load schedule.

The model used for controller design is imperfect in service. Coal quality varies across shipments, heat absorption changes as heat-transfer surfaces foul, valve and fuel-delivery characteristics drift, and unmodeled coupling becomes more visible during load following. These effects create persistent mismatch between the controller model and the true process response. A controller that appears feasible under the nominal model, whether PID, LQR, MPC, or a learned policy, can still drive the measured plant state toward a pressure, enthalpy, or power violation.

## 1.2 Safety-Filter View

This paper studies an overlay safety filter, not a replacement for the plant controller. The filter receives the measured state $x = [r_B, p_m, h_m, N_e, \tau_f]^\top$ and the upstream controller's proposed action $u_{\text{prop}}$, checks that action against process safety constraints, and returns the nearest admissible action $u^*$. This placement suits industrial measurement and control because it separates the tracking task from the protection task: the existing controller handles load following and economic operation, while the safety layer handles last-mile constraint enforcement.

The projection mechanism is a quadratic program (QP) built from control barrier functions (CBFs) [1]. For constraints whose input appears only after repeated differentiation, high-order CBFs (HOCBFs) [2] use a recursive $\psi$-chain to enforce the correct relative degree. This structure matters in boiler-turbine operation. Pressure has relative degree two in the stabilized model, whereas enthalpy and power have relative degree one. A first-order CBF applied uniformly to all constraints is structurally mismatched and, as shown experimentally, provides no protection under model mismatch.

### 1.3 Technical Gap: Uncertainty Propagation in High-Relative-Degree Constraints

Gaussian processes (GPs) can learn model residuals from commissioning or operation data while providing posterior uncertainty [3, 4]. Existing GP-CBF methods usually apply the uncertainty bound directly to a first-order CBF constraint [5, 6, 7]. Recent GP-HOCBF work extends this idea to high-order constraints through global chance-constraint or convex reformulations [8, 9]. These studies establish useful theory, but they leave a commissioning question open for process-control use: when the residual model is learned from finite data, how much of the theoretical robustness margin should be applied in the real-time QP?

For high-relative-degree constraints, the answer is not obvious because uncertainty propagates through every level of the HOCBF $\psi$-chain. The full worst-case bound can over-tighten the QP, reduce actuator authority, and trigger fallback behavior. Using no margin can also be unsafe when the perturbation is state-dependent and amplifies along the trajectory. The practical problem is therefore to derive a margin and then calibrate its use for the perturbation structure observed during commissioning.

### 1.4 Contributions

We introduce RoCBF-Net, a commissioning-calibrated GP-HOCBF safety filter for supercritical boiler-turbine control. The paper makes four contributions:

1. **Industrial safety-filter architecture for boiler-turbine measurement and control.** RoCBF-Net is formulated as an overlay between any upstream controller and the actuators. It uses measured boiler-turbine states, correct HOCBF relative degrees, GP residual correction, and a QP projection to enforce pressure, enthalpy, and power constraints without modifying the upstream control law.

2. **Compositional uncertainty propagation through the HOCBF chain.** The method propagates GP posterior uncertainty through each HOCBF $\psi$-level to form a state-dependent margin $\epsilon(x)$. The scalar factor $\epsilon_\kappa$ then scales this margin in the

online constraint $Au \leq b - \epsilon_\kappa \epsilon(x)$, recovering the worst-case certificate at $\epsilon_\kappa = 1$ and enabling less conservative calibrated operation at smaller values.

3. **Evidence that GP mean correction, not the full worst-case margin, is the dominant safety mechanism.** On a 5th-order 1000 MW boiler-turbine benchmark with $\Phi$-scaled model mismatch, GP mean correction reduces HOCBF violations by more than 500 times and achieves $\leq 0.18\%$ violation in six of seven uncertainty scenarios. The full theoretical margin $\epsilon_\kappa = 1.0$ is not the preferred operating point in any tested condition.

4. **Deployment envelope for robustness-margin calibration.** A sensitivity sweep shows that $\epsilon_\kappa = 0$ is sufficient for additive perturbations in the tested regime, $\epsilon_\kappa = 0.1$ restores zero violation under moderate state-dependent coupling, and positive margins become harmful beyond $\gamma \geq 1.5$. The result is a practical schedule for selecting $\epsilon_\kappa$ during commissioning, with diagnostics for leaving the calibrated envelope.

# 2 Related Work

## 2.1 Industrial Process Control Under Model Mismatch

Industrial process-control systems usually separate tracking, protection, and actuator coordination across several software and hardware layers. For boiler-turbine operation, a new method is most useful when it can be inserted into the existing measurement and control loop without replacing the coordinated controller. This study therefore treats RoCBF-Net as a supervisory safety filter rather than another standalone controller. The evaluation is tied to a specific boiler-turbine loop, with measured variables, process constraints, disturbance scenarios, and computational latency reported explicitly.

## 2.2 Constrained Control and Safety Filtering

MPC is the dominant constrained control methodology in process industries [10, 11], and offset-free variants address steady-state model mismatch [12]. Recent boiler-turbine MPC work in the SAGE measurement and control literature still treats nonlinear feasibility, input constraints, and load-response capability as central design challenges [13]. Cautious MPC [14] incorporates GP uncertainty into the prediction horizon and provides chance-constrained operation at increased computational cost. Here, NMPC is used as a strong benchmark because it directly optimizes over a short horizon. RoCBF-Net instead performs a one-step projection intended for low-latency safety intervention.

Control barrier functions enforce forward invariance through QP-based control projection [1, 15]. HOCBFs [2] extend this idea to high-relative-degree constraints. Robust

CBF formulations [16, 17] add margins based on structural disturbance bounds. These margins are often applied at their full theoretical value. The present results show why this choice can be counterproductive in a process-control QP: $\epsilon_\kappa = 1.0$ over-constrains the feasible set in the tested benchmark conditions, whereas smaller calibrated values preserve both safety and control authority.

## 2.3 GP-Enhanced CBFs

GPs provide nonparametric residual learning with uncertainty quantification [3, 4]. Several GP-CBF methods apply posterior uncertainty to first-order CBF constraints [5, 6, 7]. Aali and Liu [8] extend GP-HOCBF integration to high-order constraints through a second-order cone reformulation, and Zhang et al. [9] develop an adaptive sparse-GP HOCBF filter for robot control. Lederer et al. [18] provide uniform GP error bounds that are complementary to this line of work.

RoCBF-Net differs in two ways. First, uncertainty is propagated recursively through the HOCBF $\psi$-chain rather than applied only as a top-level aggregate. Second, the margin is treated as a commissioning parameter. This distinction is central in boiler-turbine operation, where a theoretically valid margin can still make the real-time QP infeasible when actuator authority is limited.

## 2.4 Safety Margin Calibration

The safety-filter literature often treats the robustness margin as a certificate requirement rather than an operating parameter. Industrial commissioning, in contrast, routinely selects gains, thresholds, and protection margins from process tests. This paper brings that practice into GP-HOCBF filtering by introducing $\epsilon_\kappa$ as a tunable factor. The experiments identify three regimes: GP mean correction is sufficient for additive uncertainty; a small positive margin is needed for moderate state-dependent coupling; and beyond the deployment envelope, increasing the margin worsens feasibility and should trigger fallback rather than tighter filtering.

In the broader safe-control literature, shielding approaches [19] correct policy actions, constrained reinforcement learning methods [20] enforce expected constraints, and adaptive CBFs [21] estimate uncertain parameters online. RoCBF-Net is closest to shielding in architecture, but its intervention is model-based and constraint-specific. It returns the nearest action satisfying pressure, enthalpy, and power HOCBF constraints under a calibrated GP residual model.

# 3 RoCBF-Net Safety-Filter Architecture

## 3.1 Industrial Safety-Filter Workflow

This section first describes the online safety-filter implementation and then gives the probabilistic margin certificate that supports it. The experiments in Section 4 determine how much of that margin should be applied under each mismatch regime.

Figure 1 summarizes the overlay architecture. The upstream controller produces a deviation command $v_{\text{prop}}$ around the LQR-stabilized equilibrium controller. The safety filter then returns the closest command $v^*$ that satisfies the process safety constraints. The total actuator command applied to the boiler-turbine model is

$$u = u_0 + K(x_0 - x) + v^*, \tag{1}$$

where $u = [u_B, D_{fw}, u_t]^\top$ denotes coal feed, feedwater flow, and turbine-valve opening. This deviation-control form is used because the CCS dynamics are stiff. The HOCBF and QP operate on the stabilized linearized dynamics, while evaluation rollouts use the nonlinear 5th-order plant with $\Phi(p_m)$-scaled control effectiveness.

**Online filtering algorithm.** At each sampling instant, RoCBF-Net executes the following steps:

**Step 1:** Read the measured state $x_k$ and obtain the proposed deviation action $v_{\text{prop},k}$ from the upstream controller.

**Step 2:** Evaluate the GP residual model at $x_k$ to obtain the mean correction $\mu_{\text{GP}}(x_k)$ and posterior standard deviations $\sigma_{\text{GP}}(x_k)$.

**Step 3:** Build the mean-corrected HOCBF constraints using the correct relative degree for each process limit.

**Step 4:** Propagate GP posterior uncertainty through the HOCBF $\psi$-chain to compute the compositional margin $\epsilon(x_k)$.

**Step 5:** Solve the real-time QP

$$v_k^* = \arg\min_v \|v - v_{\text{prop},k}\|_2^2 \quad \text{s.t.} \quad A(x_k)v \le b(x_k) - \epsilon_\kappa \epsilon(x_k), \tag{2}$$

with actuator box constraints and row scaling.

**Step 6:** Apply $u_k = u_0 + K(x_0 - x_k) + v_k^*$ to the plant, and log intervention, feasibility, and constraint-margin diagnostics.

Table 1: Measurement/control variables and implementation settings used by the RoCBF-Net safety filter on the 5th-order CCS benchmark.

| Item | Setting | Role in the filter |
|---|---|---|
| Measured state | $[r_B, p_m, h_m, N_e, \tau_f]^\top$ | Boiler gas state, separator pressure, enthalpy, turbine power, and fuel-delay state used for CBF and GP evaluation. |
| Manipulated inputs | $[u_B, D_{fw}, u_t]^\top$ | Coal feed, feedwater flow, and turbine valve opening; QP optimizes the deviation command $v$. |
| Pressure limit | 13–24 MPa, $m = 2$ | High-order pressure CBF with two $\psi$-levels. |
| Enthalpy limit | 2670–2830 kJ kg$^{-1}$, $m = 1$ | Active heat-absorption safety constraint. |
| Power limit | $1000 \pm 50$ MW, $m = 1$ | Load-following and turbine-output safety constraint. |
| Sampling and hold | $T_s = 1$ s, zero-order hold | Online QP solved once per measurement sample. |
| GP calibration | 500 samples per scenario, Matérn-5/2 residual GPs | Learns $\Delta f$ mean correction and posterior uncertainty. |
| QP implementation | qpax primal solve, row scaling, box constraints | Projects $v_{\text{prop}}$ to the nearest admissible $v^*$. |
| Fallback behavior | Slack relaxation and LQR fallback | Invoked when the tightened CBF-QP is infeasible. |
| Runtime target | $\sim$25 ms per step on RTX 4090 GPU | Fits within the 1 s sampling budget of the benchmark loop. |

**Operating-point interpretation.** RoCBF-Net denotes the full safety-filter architecture in Fig. 1: HOCBF constraints with the correct relative degrees, GP residual mean correction, compositional uncertainty propagation, and a tunable scalar $\epsilon_\kappa$. The experimental rows labelled GP-HOCBF, RHOCBF, and online GP-RHOCBF are fixed operating points or ablations of this architecture, not separate proposed methods. Thus, $\epsilon_\kappa = 0$ tests mean correction without an additional uncertainty margin, $\epsilon_\kappa = 1$ tests the full worst-case margin, and the recommended operating point is selected from the commissioning schedule in Secs. 4.3–4.4.

### 3.2 Problem Formulation

Consider a control-affine system with model mismatch:

$$\dot{x} = \underbrace{f_0(x) + g_0(x)u}_{\text{nominal}} + \underbrace{\Delta f(x)}_{\text{residual}}, \quad x \in \mathbb{R}^n,\ u \in \mathcal{U} \subset \mathbb{R}^m \tag{3}$$

where $\Delta f$ represents unmodeled dynamics, parameter drift, and external disturbances. We assume that the input matrix is known ($\Delta g = 0$), consistent with settings in which actuator characteristics are well characterized. In scenarios where actuator degradation mainly affects process variables through reduced throughput (for example, valve fouling reducing steam extraction efficiency) rather than through actuator-gain changes, the dominant effect is captured by $\Delta f$. The S5 benchmark scenario models actuator degradation in this form.

For a constraint $h(x) \geq 0$ with relative degree $m$, the HOCBF $\psi$-chain is:

$$\psi_0 = h(x), \quad \psi_i = \dot{\psi}_{i-1} + \alpha_i(\psi_{i-1}), \ \ i = 1, \dots, m \tag{4}$$

With linear $\alpha_i(r) = k_i r$, the constraint $A_0(x)u \leq b_0(x)$ enforces forward invariance when the model is exact.

### 3.3 GP Residual Model

Each component of $\Delta f_j$ is modeled as an independent GP with a Matérn-5/2 kernel. Given $N$ training points, the posterior provides mean $\mu_{\mathrm{GP},j}(x)$ and variance $\sigma^2_{\mathrm{GP},j}(x)$. The GP-UCB bound gives:

$$|\Delta f_j(x) - \mu_{\mathrm{GP},j}(x)| \leq \beta \sigma_{\mathrm{GP},j}(x), \quad \text{w.p.} \geq 1 - \delta/n \tag{5}$$

with $\beta = \sqrt{2(\gamma_N + 1 + \ln(n/\delta))}$ and maximum information gain $\gamma_N$ [4].

We define the mean-corrected drift $\hat{f} = f_0 + \mu_{\mathrm{GP}}$ and posterior residual $\Delta\hat{f} = \Delta f - \mu_{\mathrm{GP}}$.

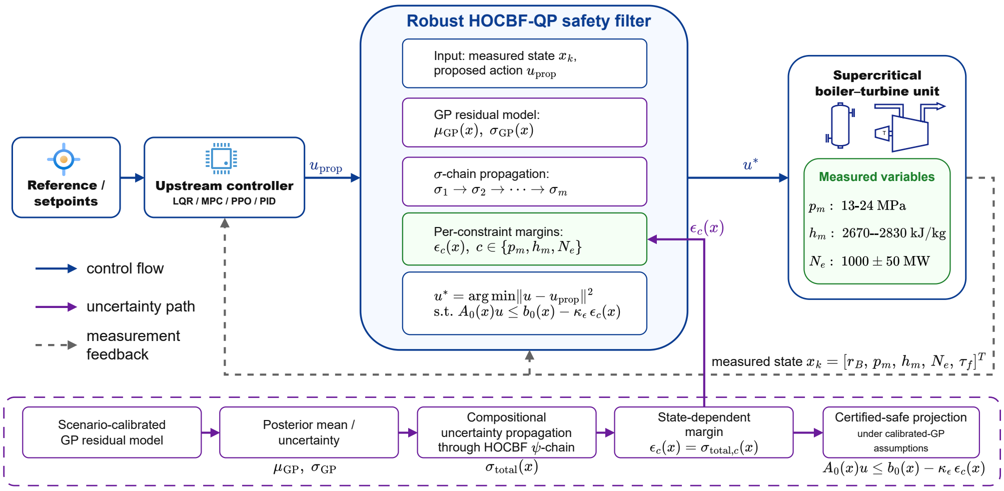


Figure 1: RoCBF-Net safety filtering architecture. The upstream controller (LQR, MPC, PID, or learned policy) proposes an action $u_{\text{prop}}$. The filter evaluates it against current plant measurements and the GP residual model, then projects it to the nearest certified-safe action $u^*$ through the QP constraint $Au \le b - \epsilon_\kappa \epsilon(x)$. HOCBF relative degrees, GP calibration mode, and $\epsilon_\kappa$ are independently configurable.

### 3.4 Compositional Robustness Margin

The perturbation at level $i$ is $\delta_i = \psi_i - \hat{\psi}_i^0$, satisfying:

$$\delta_1 = L_{\Delta \hat{f}} h \tag{6}$$
$$\delta_i = L_{\Delta \hat{f}} \hat{\psi}_{i-1}^0 + (L_{\hat{f}} + k_i)\delta_{i-1} + L_{\Delta \hat{f}} \delta_{i-1},\ i \geq 2 \tag{7}$$

The compositional margin is constructed as:

$$\sigma_1(x) = \beta \sum_j |\partial h / \partial x_j| \sigma_{\mathrm{GP},j}(x) \tag{8}$$
$$\sigma_i(x) = \beta \sum_j |\partial \hat{\psi}_{i-1}^0 / \partial x_j| \sigma_{\mathrm{GP},j}(x) + (\|L_{\hat{f}}\|_{\mathrm{op}} + k_i)\sigma_{i-1}(x) + \sigma_{\mathrm{cross}}^{(i)}(x) \tag{9}$$
$$\sigma_{\mathrm{total}}(x) = \sigma_m(x) + \sum_{j=1}^{m-1} c_j \sigma_j(x) + \sigma_{\mathrm{ctrl}}(x) \tag{10}$$

Here, $\|L_{\hat{f}}\|_{\mathrm{op}}$ is the operator norm (the spectral norm of the Jacobian $\partial \hat{f}/\partial x$ at the equilibrium operating point), $\sigma_{\mathrm{cross}}^{(i)}(x)$ bounds the cross term $|L_{\Delta \hat{f}} \delta_{i-1}|$ via Lemma S1, and $c_j = \prod_{i=j+1}^{m-1}(\|L_{\hat{f}}\|_{\mathrm{op}} + k_i)$ are the chain coupling weights. The resulting HOCBF constraint is $A_0(x)u \leq b_0(x) - \epsilon_\kappa \cdot \sigma_{\mathrm{total}}(x)$.

### 3.5 Theoretical Guarantee: Worst-Case Bound

**Theorem 1** (Probabilistic forward invariance)**.** *Suppose that the input matrix is exact ($\Delta g = 0$), each GP residual component satisfies the RKHS boundedness condition required by the GP-UCB confidence bound, the GP posterior is calibrated on fixed i.i.d. training data, the relative degree is known, the residual dynamics are sufficiently small and Lipschitz on the operating set, and the robust QP is feasible. For $\epsilon_\kappa = 1$, if $\epsilon(x) \geq |\Delta(x,u)|$ for all admissible $u$, then $\mathcal{C}$ is forward invariant with probability at least $1 - \delta$.*

*Proof sketch.* **Step 1 (GP calibration):** By the RKHS boundedness and GP calibration conditions together with (5), $|\Delta \hat{f}_j| \leq \beta \sigma_{\mathrm{GP},j}$ holds with probability $\geq 1 - \delta/n$ per dimension; by union bound over the $n$ state dimensions, this holds simultaneously with probability $\geq 1 - \delta$.

**Step 2 (Compositional bound):** By induction, $|\delta_i| \leq \sigma_i$. The cross term $\sigma_{\mathrm{cross}}^{(i)}$ is bounded via Lemma S1 (Appendix): $|L_{\Delta \hat{f}} \delta_{i-1}| \leq G_\delta^{(i-1)} \beta \|\sigma_{\mathrm{GP}}\|_2$.

**Step 3:** $|\Delta(x,u)| \leq \sigma_{\mathrm{total}}(x) = \epsilon(x)$ for $\epsilon_\kappa = 1$, completing the certificate. □

Theorem 1 is a *worst-case* statement. Setting $\epsilon_\kappa = 1$ applies the full compositional margin used in the certificate, so forward invariance holds with the stated probability whenever the tightened QP remains feasible under the actuator authority condition. Feasibility is not implied by the margin itself; it depends on the safe-set geometry, actuator limits, and feasible-control set $\mathcal{U}$. The guarantee is conservative because it covers the worst-case combination of GP posterior uncertainty and $\psi$-chain amplification. The next subsection treats $\epsilon_\kappa$ as a tunable parameter that recovers this bound at $\epsilon_\kappa = 1$ while permitting less conservative calibration for specific uncertainty regimes.

### 3.6 $\epsilon_\kappa$ as a Tunable Safety Factor

The robustness scaling factor $\epsilon_\kappa$ in the QP constraint $A_0(x)u \leq b_0(x) - \epsilon_\kappa \cdot \epsilon(x)$ controls the safety-margin/control-authority trade-off:

- $\epsilon_\kappa = 0$: The QP enforces $A_0 u \leq b_0$, i.e., the nominal HOCBF constraint with GP mean correction only. No robustness margin is applied beyond the mean correction itself.
- $\epsilon_\kappa = 1$: The QP enforces the full compositional margin $\epsilon(x)$, recovering the worst-case certificate of Theorem 1.
- $0 < \epsilon_\kappa < 1$: The QP enforces a fraction of the compositional margin, trading theoretical coverage for reduced QP conservatism.

The empirical question is whether the worst-case setting ($\epsilon_\kappa = 1$) is needed in the tested process-control benchmark, and whether the preferred $\epsilon_\kappa$ changes with the structure and strength of model mismatch. Section 4 shows that: (i) $\epsilon_\kappa = 0$ (GP correction only) is sufficient for additive bounded perturbations, achieving $\leq 0.18\%$ violation across six of seven uncertainty scenarios; (ii) $\epsilon_\kappa = 0.1$ is required for state-dependent coupling, recovering zero violation from 33.5% failure; (iii) $\epsilon_\kappa = 1.0$ is not the preferred choice under any tested condition; and (iv) the safe $\epsilon_\kappa$ range collapses as coupling strength increases, defining a deployment envelope that depends on the perturbation amplification rate $\gamma$.

### 3.7 Data-Driven vs. Floor-Driven $\epsilon(x)$

The GP posterior includes a regularization floor $\sigma_{\text{floor}} = 10^{-4}$. Under constant perturbations, $\sigma_{\text{GP}}$ is near-uniform and $\epsilon(x)$ is approximately constant per constraint; the margin is *floor-driven*, with the GP contributing a near-constant safety buffer. Under state-dependent perturbations (S3: Coupled), $\sigma_{\text{GP}}$ becomes heterogeneous and $\epsilon(x)$ is state-dependent; the margin is *data-driven*, reflecting the spatial structure of uncertainty. This distinction explains why additive perturbation scenarios can use $\epsilon_\kappa = 0$ in the tested regime: the GP mean correction already provides the useful adjustment, and the floor-driven $\epsilon(x)$ mainly adds QP conservatism.

### 3.8 Sampled-Data Implementation

Theorem 1 is a continuous-time result. The implementation uses discrete sampling ($T_s = 1\,\mathrm{s}$, zero-order hold). The full safety-filter pipeline latency is $\sim$25 ms on the GPU after JIT compilation, including the PPO forward pass, GP inference, and QP solve. When the QP is infeasible, the controller invokes a slack-penalised relaxation with LQR fallback. Inter-sample constraint satisfaction is checked empirically in Section 4; a formal discrete-time invariance theorem is outside the present scope.

## 4 Experimental Validation

The experiments evaluate RoCBF-Net as a real-time safety layer in an industrial measurement and control loop. We report CBF violation rates together with tracking IAE, input variation, QP intervention rate, inter-sample behavior, and per-step computational latency. These metrics characterize both safety and commissioning suitability in a DCS-like setting.

### 4.1 Benchmark and Baselines

The benchmark is a 5th-order nonlinear simulation model of a 1000 MW ultra-supercritical boiler-turbine unit with state $x = [r_B, p_m, h_m, N_e, \tau_f]^\top$ and three control inputs. Six safety constraints are enforced: main steam pressure ($m = 2$, 13–24 MPa), separator enthalpy ($m = 1$, 2670–2830 kJ/kg), and power output ($m = 1$, $\pm$50 MW). Table 1 summarizes the measured variables and implementation settings. During evaluation, the nonlinear plant uses state-dependent $\Phi(p_m)$-scaled control effectiveness while the HOCBF uses the linearized model. This mismatch stresses the safety filter without changing the controller design model.

Six static perturbation scenarios model realistic disturbances: S1 (coal quality drift, $-15\%$ heating value), S2 (feedwater pump cycling), S3 (coupled state-dependent, $\delta \propto x$), S4 (nonlinear fouling), S5 (valve degradation, modelled as equivalent drift $\Delta f$ via reduced steam extraction efficiency; the $\Delta g = 0$ assumption is maintained throughout all scenarios), and S6 (fuel quality variation). Scenarios S1–S4 are common to the 3rd- and 5th-order CCS models; S5 and S6 are specific to the 5th-order model, affecting the extended states $N_e$ and $\tau_f$.

The comparison isolates the safety layer. PPO, PPO-Lagrangian, PPO-CBF, and PPO-HOCBF use the same upstream learned action source and differ only in the post-controller constraint mechanism: no safety layer, dual-descent penalty, first-order CBF ($m$=1 for all constraints), and correct HOCBF relative degrees ($m$=$2, 1, 1$) without GP correction. The RoCBF-Net family is evaluated at fixed operating points: mean-corrected GP-HOCBF with no additional margin ($\epsilon_\kappa$=0), full-margin RHOCBF with scenario-specific GP ($\epsilon_\kappa$=1.0), and online GP-RHOCBF with uncalibrated full margin. The calibrated operating schedule is then assessed in Secs. 4.3–4.4. NMPC (SLSQP solver, $N$=5 horizon) is included as a strong optimization-based reference. GP pretraining uses $N = 500$ points per scenario, and evaluation uses 5 seeds $\times$ 50 episodes $\times$ 500 steps. The study is a simulation-benchmark validation of a pre-deployment safety-filter design and commissioning protocol. Plant historian replay, hardware-in-the-loop tests, and site-specific protection reviews remain required before field deployment.

## 4.2 Safety Performance

Table 2: Constraint violation rate (%) under $\Phi$-scaled nonlinear dynamics. 0% rates: 0/250 episodes, 95% Wilson upper bound 1.51%.

| Method | Nom. | S1 | S2 | S3 | S4 | S5 | S6 |
|---|---|---|---|---|---|---|---|
| PPO (no safety) | 0.08 | 99.92 | 99.53 | 99.59 | 99.05 | 99.12 | 99.83 |
| PPO-Lagrangian | 0.08 | 99.92 | 99.53 | 99.59 | 99.05 | 99.12 | 99.83 |
| PPO-CBF (1st-order, $m$=1) | 0.08 | 99.92 | 99.53 | 99.59 | 99.05 | 99.12 | 99.83 |
| PPO-HOCBF ($m$=2, 1, 1, no GP) | 0.00 | 99.92 | 99.53 | 99.59 | 99.05 | 99.12 | 99.83 |
| RoCBF-Net mean-only ($\epsilon_\kappa$=0) | 0.00 | 0.12 | 0.04 | 39.65 | 0.04 | 0.15 | 0.18 |
| RoCBF-Net full margin ($\epsilon_\kappa$=1.0) | 0.00 | 1.38 | 99.43 | 16.96 | 32.19 | 0.00 | 0.00 |
| Online RoCBF-Net full margin (uncal.) | 0.00 | 15.02 | 99.40 | 48.58 | 49.16 | 0.00 | 0.00 |
| NMPC ($N$=5, SLSQP) | 0.00 | 0.08 | 0.08 | 0.08 | 0.04 | 0.08 | 0.08 |

Table 2 reports the main safety results. A nominal safety layer without residual learning fails under model mismatch. PPO without safety, PPO-Lagrangian, and PPO-CBF all exceed 99% violation under any dynamics-affecting scenario. PPO-HOCBF with correct relative degrees achieves 0.00% on nominal dynamics but still exceeds 99% under uncertainty. Correct HOCBF structure is therefore necessary but insufficient without uncertainty modeling.

GP mean correction is the dominant safety mechanism. RoCBF-Net at $\epsilon_\kappa$=0 reduces violation to $\leq$0.18% under six of seven uncertainty scenarios, a >500$\times$ improvement over PPO-HOCBF, without applying an additional robustness margin. The exception is S3 (Coupled, state-dependent), where GP mean correction alone leaves 39.65% violation because the perturbation includes feedback amplification.

The full theoretical margin tests the conservative end of the certificate, not the recommended operating point. RoCBF-Net with $\epsilon_\kappa$=1.0 degrades on S2 (99.43%) and S4 (32.19%) compared with $\epsilon_\kappa$=0 (0.04% and 0.04%, respectively). The margin tightens the QP feasible set, triggers constraint-dropping fallback, and can increase violations. The sensitivity sweep in Sec. 4.3 is used to select the calibrated $\epsilon_\kappa$ schedule.

NMPC with $N$=5 horizon achieves $\leq$0.08% violation across all conditions and serves as the strongest optimization-based safety reference in this study. Its per-step cost ($\sim$254 ms, CPU) is approximately 10× higher than the implemented CBF-QP pipeline (∼25 ms, GPU), a hardware- and implementation-dependent trade-off discussed in Sec. 4.5.

The remaining violations are enthalpy-dominated. Per-constraint logs show zero observed violations on pressure ($m$=2) and power ($m$=1); the aggregate violations arise from the enthalpy constraint ($m$=1). This bottleneck reflects its tight safety bounds (2670–2830 kJ/kg) and the direct impact of heat-absorption uncertainty on the enthalpy state.

These results establish the mechanism hierarchy: correct HOCBF structure is necessary, GP uncertainty correction is the primary safety mechanism, and $\epsilon_\kappa$ provides a tunable safety–performance trade-off that should be calibrated by uncertainty type rather than fixed at the worst-case theoretical bound.

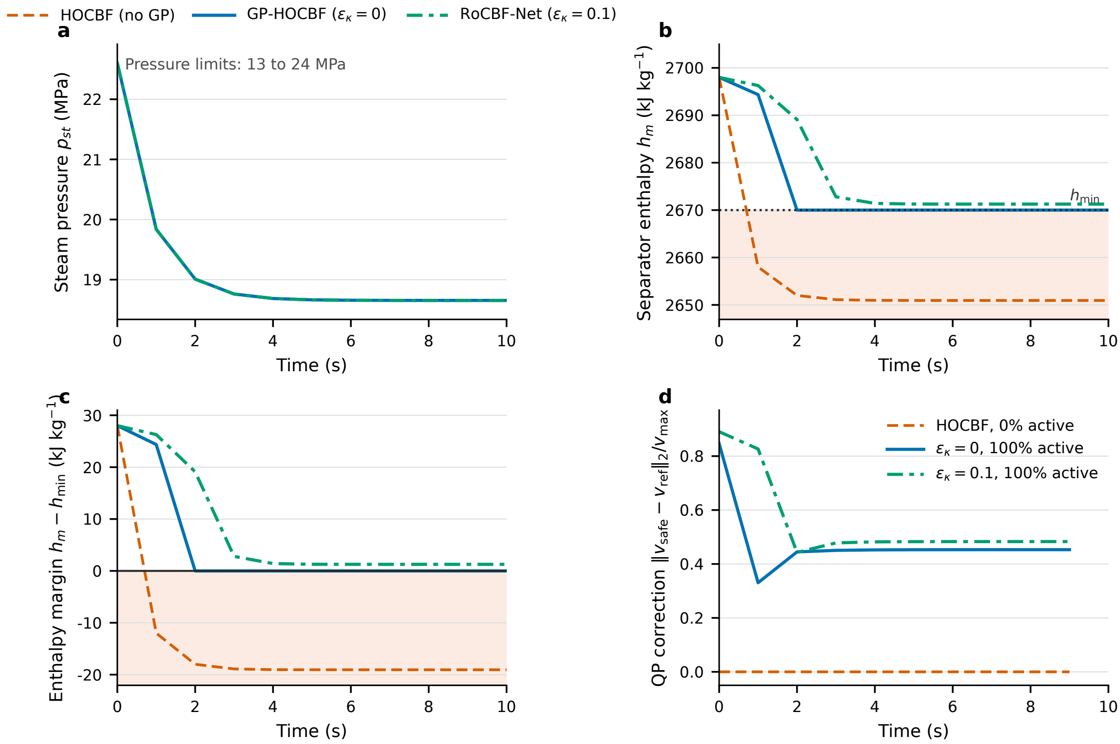


Figure 2: Process response and safety-filter intervention under the S3 coupled perturbation. **(a)** Steam-pressure response. **(b)** Separator-enthalpy response with the active lower safety limit. **(c)** Enthalpy safety margin. **(d)** Normalized QP correction magnitude. The curves show the first 10 s of a 300-step rollout; intervention percentages are computed over the full rollout.

Figure 2 translates the aggregate violation rates into process-control signals. The pressure response remains within the 13–24 MPa band for all three filters, whereas the enthalpy response separates the safety mechanisms. HOCBF without GP correction crosses the lower enthalpy bound immediately and settles outside the admissible set. GP mean correction drives the trajectory back to the boundary but leaves a short transient violation. Adding the calibrated $\epsilon_\kappa$=0.1 margin keeps enthalpy inside the safe band. Because the upstream action in this diagnostic is the zero-deviation action around the LQR-stabilized operating point, the QP correction in Fig. 2(d) is the direct intervention required to maintain the enthalpy constraint under S3.

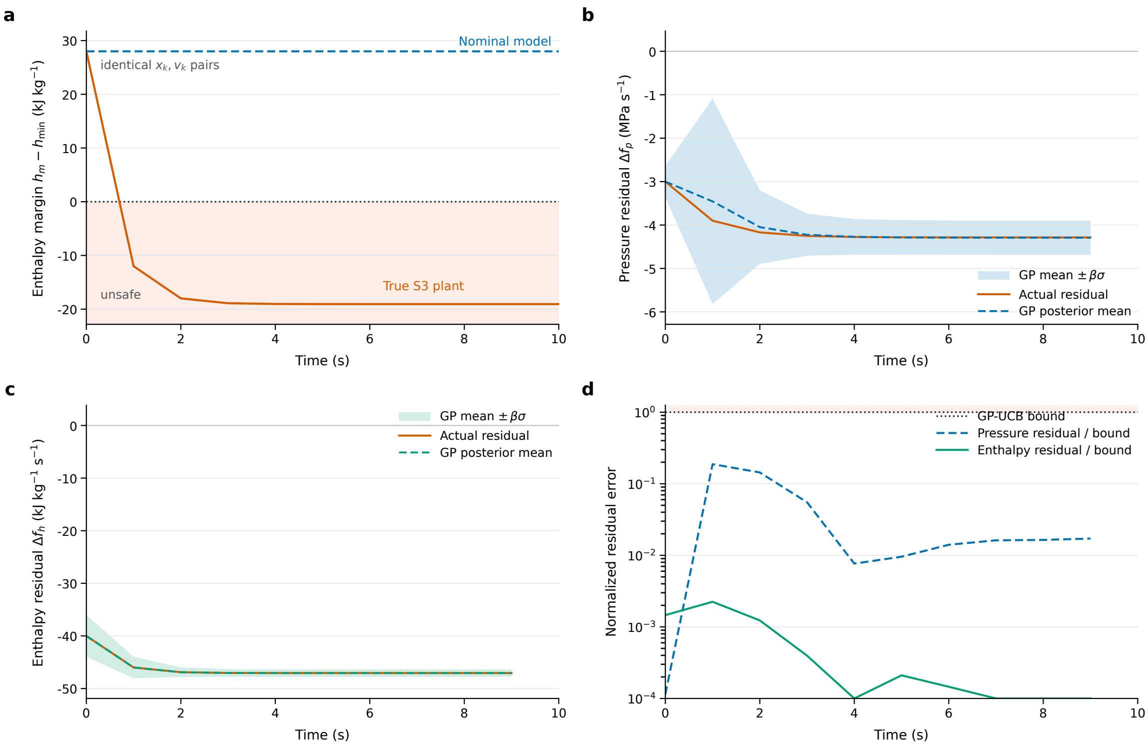


Figure 3: Direct model-mismatch diagnosis under the S3 coupled perturbation. **(a)** Nominal and true one-step enthalpy-margin responses from identical state-input pairs. **(b,c)** Pressure and enthalpy residuals compared with the scenario-specific GP posterior mean and GP-UCB envelope ($\mu_{\text{GP}} \pm \beta\sigma_{\text{GP}}$, $\beta = 35.2$, $N = 500$). **(d)** Normalized posterior residual error; values below one lie inside the GP-UCB envelope used by the robust HOCBF margin.

Figure 3 makes the model mismatch explicit. For the same state and zero-deviation input, the nominal model predicts a next-step enthalpy margin of 28.0 kJ kg$^{-1}$, whereas the true S3 plant drops to $-12.0$ kJ kg$^{-1}$. The pressure and enthalpy residual panels show that this discrepancy is structured and learnable: the GP posterior mean tracks the realized model gap, and the remaining residual stays inside the calibrated GP-UCB envelope. This mechanism explains the improvement from nominal HOCBF to GP-corrected and robust HOCBF filtering in Fig. 2.

### 4.3 $\epsilon_\kappa$ Sensitivity: One Safety Factor Does Not Fit All

To quantify the practical safety–performance trade-off, we swept $\epsilon_\kappa \in \{0, 0.1, 0.3, 0.5, 1.0\}$ across three representative 5th-order CCS conditions: S2 (additive oscillation), S3 (state-dependent coupled), and S4 (nonlinear fouling). Each setting uses 2–3 seeds. This sweep

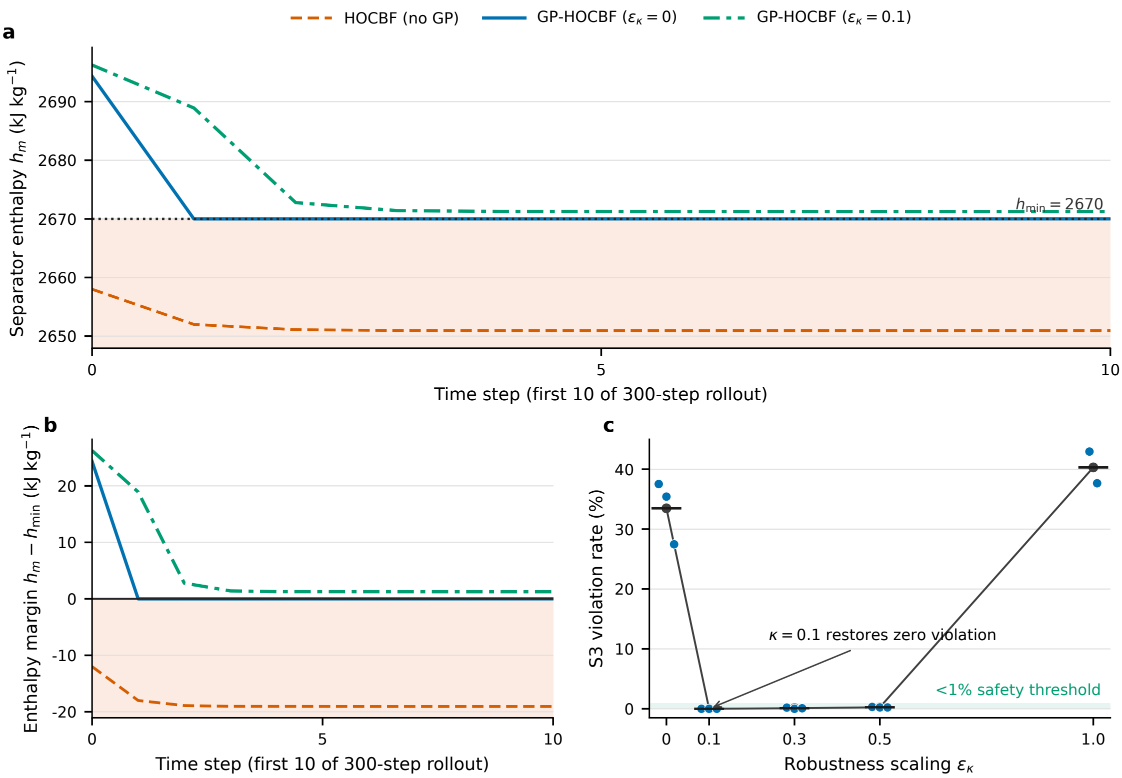


Figure 4: Failure mechanism and margin calibration under S3: Coupled perturbation. **(a)** Separator-enthalpy trajectories over the first 10 steps of a 300-step rollout for HOCBF without GP, mean-corrected GP-HOCBF ($\epsilon_\kappa$=0), and GP-HOCBF with a small margin ($\epsilon_\kappa$=0.1). The shaded region marks violation of $h_{\min} = 2670\,\mathrm{kJ\,kg^{-1}}$. **(b)** Equivalent enthalpy safety margin $h_m - h_{\min}$. **(c)** Seed-level S3 $\epsilon_\kappa$ sweep over full evaluation rollouts (dots = seeds, black ticks/line = mean; $n = 3$ except $\epsilon_\kappa$=1.0, where $n = 2$).

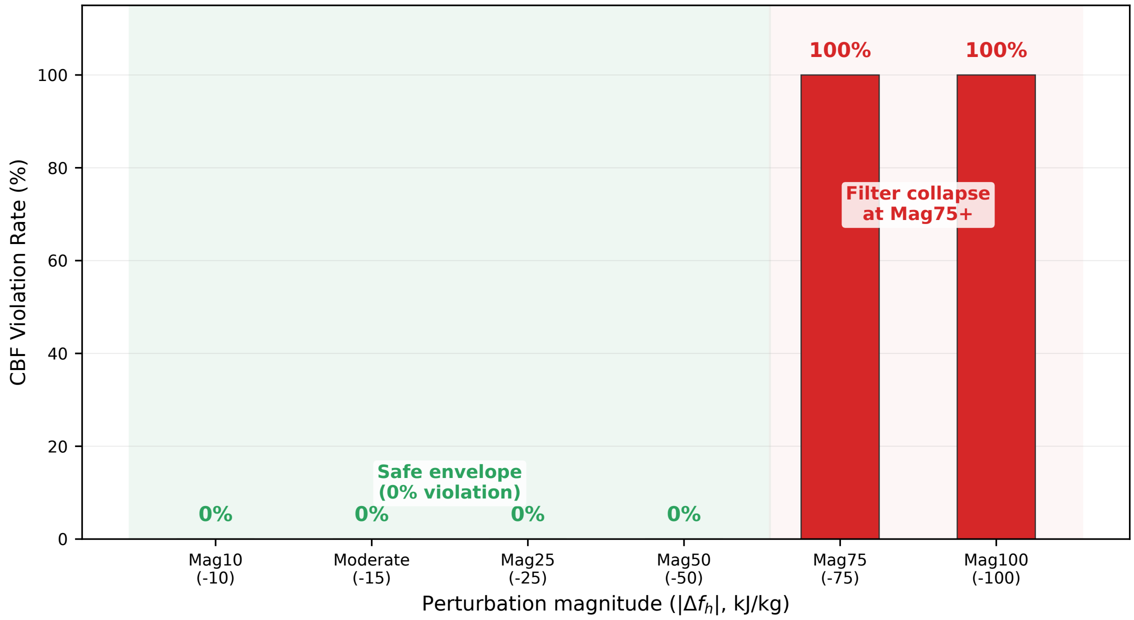


Figure 5: Safety envelope under varying S1 heat-absorption magnitude for GP-HOCBF with $\epsilon_\kappa$=0. The filter maintains zero violation up to Mag50; Mag75 and Mag100 exceed the GP correction capacity and produce 100% violation.

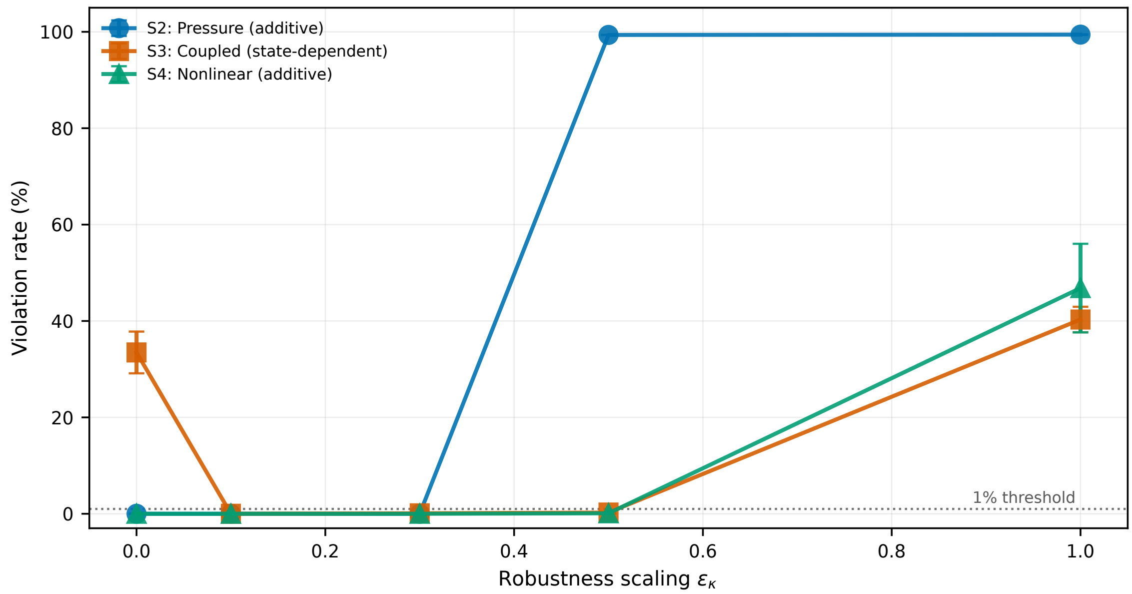


Figure 6: $\epsilon_\kappa$ sensitivity across three representative uncertainty scenarios. Additive uncertainty (S2, S4) is safest at $\kappa$=0; state-dependent uncertainty (S3) requires a positive but non-maximal margin.

is a commissioning diagnostic, not a replacement for the 5-seed safety table; the Supplemental Material reports the seed counts behind each cell.

**Additive uncertainty (S2, S4):** $\kappa$=0 achieves zero violation. GP mean correction alone compensates for bounded, non-amplifying perturbations. Increasing $\kappa$ to 0.5 collapses QP feasibility on S2 (99.4% violation), while S4 remains safe (0.10%). The nonlinear fouling perturbation preserves a larger feasible set than the pressure oscillation. $\kappa^*_{\mathrm{S2}}$=0, $\kappa^*_{\mathrm{S4}}$=0.

**State-dependent uncertainty (S3):** $\kappa$=0 fails (33.5% violation) because GP mean correction cannot fully track the feedback-amplified perturbation. A small margin ($\kappa$=0.1) restores zero-violation safety, and safety is maintained through $\kappa$=0.5 (0.25%). The full margin $\kappa$=1.0 collapses feasibility (40.3% violation). $\kappa^*_{\mathrm{S3}}$=0.1.

**Practical recommendations:** For additive uncertainty ($|\delta|$ bounded), use the smallest margin that maintains residual coverage; in the tested additive scenarios, $\epsilon_\kappa$=0 is preferred and values up to 0.3 remain safe in the diagnostic sweep. For state-dependent uncertainty ($\delta \propto x$), $\epsilon_\kappa \in [0.1, 0.5]$ is recommended only within the tested coupling envelope. The margin is useful but should not be maximal. These findings motivate treating $\epsilon_\kappa$ as a tunable safety factor calibrated by uncertainty type rather than as a fixed theoretical constant.

## 4.4 Deployment Envelope: $\epsilon_\kappa$ Under Varying Coupling Strength

Section 4.3 established that the preferred $\epsilon_\kappa$ differs between additive and state-dependent uncertainty. The next question is whether the *strength* of state-dependent coupling also matters. We therefore swept $\epsilon_\kappa \in \{0, 0.1, 0.3, 0.5, 1.0\}$ across four S3 coupling strengths, scaling the state-dependent feedback gain $\gamma \in \{0.5, 1.0, 1.5, 2.0\}$ in the perturbation function $\delta(x) = [0, \gamma\, c_p(x_1 - x_1^0) - d_p, 0.5\gamma\, c_h(x_2 - x_2^0) - d_h, 0, 0]^\top$ (2 seeds × 50 episodes × 500 steps, 5th-order CCS). This gradient is a boundary-finding diagnostic and should be interpreted together with the 5-seed main table and the full data in the Supplemental Material.

Three regimes emerge from the coupling gradient. In the GP-sufficient regime ($\gamma < 0.5$), GP mean correction alone provides near-perfect safety because the state-dependent perturbation is weak enough for the GP prediction to track without a robustness margin.

In the $\epsilon_\kappa$-useful regime ($0.5 \le \gamma < 1.5$), the perturbation outgrows the GP correction at $\kappa$=0, and a small positive margin ($\kappa \in [0.1, 0.3]$) restores zero-violation safety. The safe $\kappa$ range narrows as $\gamma$ increases: $\kappa \in [0.1, 1.0]$ at $\gamma$=0.5 and $\kappa \in [0.1, 0.5]$ at $\gamma$=1.0.

At the deployment boundary ($\gamma \ge 1.5$), the perturbation amplification rate exceeds the CBF Lie-derivative chain's compensation capacity. Adding $\epsilon_\kappa$ constrains the QP without addressing that structural limitation. $\kappa$=0 remains marginally safe (0.51% at $\gamma$=1.5, 0.19% at $\gamma$=2.0), but any positive $\kappa$ collapses feasibility (75–98% at $\gamma$=1.5, 98–

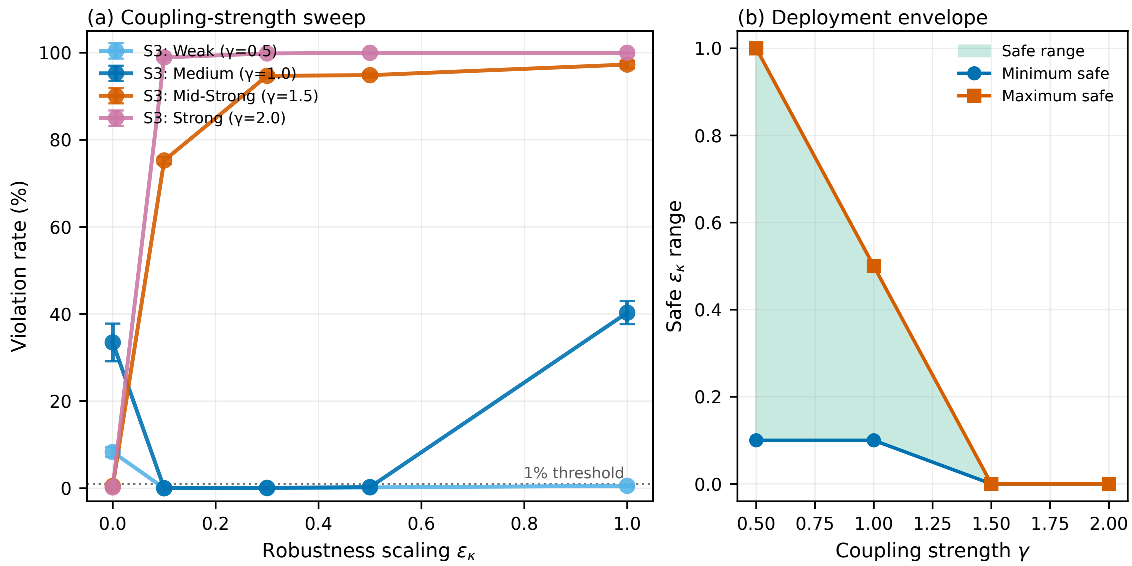


Figure 7: Deployment envelope of $\epsilon_\kappa$ under varying S3 coupling strength $\gamma$ (5th-order CCS). **Left:** violation rate vs. $\epsilon_\kappa$ for four coupling strengths. **Right:** safe $\epsilon_\kappa$ range, defined as <1% violation. A small positive margin is useful at $\gamma = 0.5$ and $\gamma = 1.0$, but the envelope closes between $\gamma = 1.0$ and $\gamma = 1.5$. At $\gamma \geq 1.5$, positive margins mainly collapse QP feasibility.

100% at $\gamma$=2.0). In this regime, QP infeasibility is a diagnostic signal: the unit should fall back to a conservative schedule or raise an alarm rather than increase the margin.

Table 3: Deployment envelope of $\epsilon_\kappa$ as a function of S3 coupling strength $\gamma$. Safe range defined as achieving <1% violation rate. 5th-order CCS, 2 seeds × 50 episodes × 500 steps.

| $\gamma$ | $\kappa^*$ | Safe $\kappa$ Range | Min Violation | Regime |
|---|---|---|---|---|
| 0.5 (Weak) | 0.1 | $[0.1, 1.0]$ | 0.00% | $\epsilon_\kappa$ useful |
| 1.0 (Medium) | 0.1 | $[0.1, 0.5]$ | 0.00% | $\epsilon_\kappa$ useful |
| 1.5 (Mid-strong) | 0.0 | $\{0\}$ | 0.51% | Deployment boundary |
| 2.0 (Strong) | 0.0 | $\{0\}$ | 0.19% | Deployment boundary |

**Implication.** The coupling gradient defines a deployment envelope for $\epsilon_\kappa$ (Table 3). Commissioning should characterize the expected coupling strength and then select $\kappa$ from the corresponding regime, rather than fixing $\kappa$ to the theoretical worst-case value. For units whose replay data match the S3-like medium-coupling profile ($\gamma \approx 1.0$), $\kappa = 0.1$ gives the best safety–performance trade-off in this benchmark. For units entering service with uncertain perturbation characteristics, $\kappa = 0.3$ should be used only after diagnostic tests indicate that the unit is not near the $\gamma \geq 1.5$ boundary. At that boundary, no tested $\epsilon_\kappa$ value improves safety; QP infeasibility should trigger fallback rather than a margin increase.

## 4.5 Filter Diagnostics and Computational Performance

Figure 4 isolates the S3 enthalpy bottleneck. Panels (a,b) show the first 10 steps of a 300-step rollout because boundary crossing occurs immediately. HOCBF without GP drops below the lower enthalpy bound, GP mean correction removes most of the offset but leaves the trajectory close to the active boundary, and a small positive margin ($\epsilon_\kappa$=0.1) restores zero violation on the full rollout. The seed-level sweep in Fig. 4(c) connects this trajectory-level mechanism to the aggregate S3 result: $\epsilon_\kappa$=0 leaves 33.47% mean violation in the kappa sweep and 39.65% in the main 5-seed table, while $\epsilon_\kappa$=0.1 restores 0% violation in the sweep.

Figure 8 quantifies the compositional $\epsilon(x)$ margin under S3. The enthalpy margin $\epsilon_h$ varies by more than one order of magnitude across the operating enthalpy range (CV 105%). This state dependence is pronounced only under state-dependent perturbations. Under constant perturbations (S1, S2, S4–S6), $\epsilon(x)$ is approximately constant and dominated by the GP regularization floor, consistent with the $\epsilon_\kappa$ sweep result that additive uncertainty requires negligible margin in the tested regime.

Figure 5 shows violation rate as a function of perturbation magnitude under GP-HOCBF ($\epsilon_\kappa$=0). The filter maintains zero violation up to Mag50 (50 kJ/kg enthalpy perturbation). At Mag75 (75 kJ/kg) and above, the perturbation exceeds the GP correction capacity and violations rise to 100%. This defines the safe operating envelope for the GP-only configuration.

Per-step computation is reported for the implemented research code, not as a hardware-normalized solver benchmark. The safety-filter pipeline (PPO+GP+QP) requires ∼25 ms after GPU JIT compilation (p95: 27 ms), including the PPO forward pass, GP inference, and QP solve. The NMPC reference (CPU, scipy SLSQP) averages 254 ms. The implemented filter therefore fits within the 1 s sampling budget and is faster than the implemented SLSQP baseline. Optimized industrial MPC packages may have different timing. For CPU-only deployment, the distilled policy achieves 1.8 ms inference, but this is an empirical policy surrogate without the certified robustness-margin calculation.

The GP-HOCBF filter is minimally invasive when the proposed action already satisfies the CBF constraints: the QP returns the action unchanged. Under $\epsilon_\kappa$=0, the QP rarely intervenes on nominal and additive-uncertainty conditions (S1, S2, S4–S6, Load Following), acting mainly when the GP residual prediction indicates an impending boundary crossing. Under S3, where GP mean correction alone is insufficient, a modest margin ($\epsilon_\kappa$=0.1) restores safety at negligible control cost. Under $\epsilon_\kappa$=1.0, the filter intervenes on nearly every step, and the QP frequently relaxes constraints through the weak-authority fallback. This confirms that $\epsilon_\kappa$ controls the intervention–safety trade-off: zero for minimal intervention on well-modeled systems, a small positive value for state-dependent uncertainty, and the full theoretical margin only when the tightened QP remains feasible.

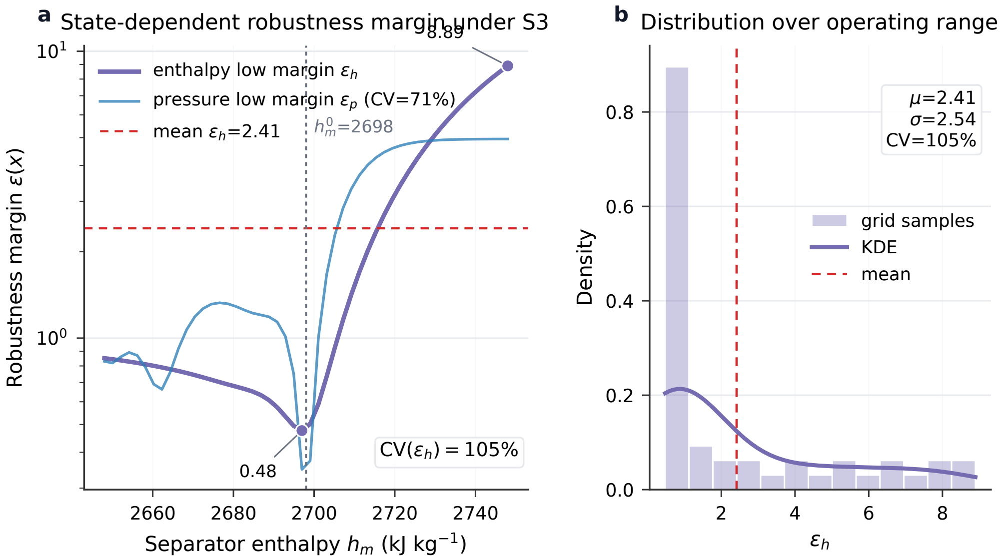


Figure 8: State-dependent enthalpy robustness margin $\epsilon_h(x)$ under S3 (Coupled, state-dependent perturbation). Left: $\epsilon_h$ (enthalpy low margin, log scale) evaluated over the operating enthalpy range $h_m^0 \pm 50\,\text{kJ/kg}$ under Φ-scaled nonlinear dynamics with scenario-specific GP. The margin varies by more than one order of magnitude, reflecting the sensitivity of $\sigma_{\text{GP}}$ to the perturbation structure (CV 105%). Right: distribution of $\epsilon_h$ over the operating range, centred at the equilibrium enthalpy $h_m^0 = 2698\,\text{kJ/kg}$. The pressure low margin $\epsilon_p$ exhibits CV 71% over the corresponding operating sweep. Under constant perturbations (S1, S2, S4–S6), $\epsilon(x)$ is approximately constant per constraint and dominated by the GP regularization floor.

Table 4: Process control performance of the mean-only RoCBF-Net operating point ($\epsilon_\kappa$=0) under Φ-scaled 5th-order CCS. Mean±std over 5 seeds × 50 episodes × 500 steps.

| Metric | S1: Heat | S3: Coupled | S5: Valve |
|---|---|---|---|
| Violation (%) | 0.12±0.11 | 39.65±9.69 | 0.15±0.08 |
| Control cost | 6560±8 | 4010±7 | 4212±8 |
| $\text{RMSE}_\text{p}$ (MPa) | 4.57±0.00 | 3.90±0.00 | 3.65±0.00 |
| $\text{RMSE}_\text{h}$ (kJ/kg) | 27.94±0.00 | 27.92±0.00 | 27.93±0.00 |
| $\text{RMSE}_\text{Ne}$ (MW) | 0.23±0.00 | 0.23±0.00 | 19.88±0.00 |

Table 4 reports process-level metrics for the mean-only operating point ($\epsilon_\kappa$=0), which is recommended for additive uncertainty scenarios (Sec. 4.3). The filter maintains consistent tracking across the three constraint-active perturbation scenarios: pressure RMSE remains within 3.65–4.57 MPa, enthalpy RMSE within 27.92–27.94 kJ/kg, and power RMSE is negligible on S1 and S3 (0.23 MW), with the expected increase on the 5th-order-specific S5 valve degradation scenario. Control cost remains within 4010–6560, indicating that GP mean correction does not impose an excessive control penalty.

The S3 column shows the limitation of $\epsilon_\kappa$=0 under state-dependent uncertainty (39.65% violation), motivating the tuned $\kappa \in [0.1, 0.5]$ recommendation from the sensitivity sweep. In that sweep, $\epsilon_\kappa$=0.1 achieves zero violation on S3 with comparable control cost.

## 4.6 Deployment Considerations

RoCBF-Net is intended as a **commissioning-calibrated safety filter** with a tunable robustness factor $\epsilon_\kappa$. Pre-deployment commissioning requires four decisions: how to train the GP, how to validate the filter before online use, how to set $\epsilon_\kappa$, and when to trigger fallback. Table 5 summarizes the corresponding protocol.

Table 5: Commissioning protocol for deploying RoCBF-Net as a safety-filter overlay.

| Decision point | Diagnostic signal | Recommended action |
|---|---|---|
| GP calibration | Held-out residual coverage and scenario-representative samples | Use a scenario-specific GP when commissioning data are available; use online adaptation only with transient monitoring. |
| Pre-deployment validation | Historian replay, hardware-in-the-loop checks, and protection-limit review | Enable online intervention only after the GP, QP fallback, and alarm logic pass offline and supervised commissioning tests. |
| Margin selection | CBF violation, QP intervention, and infeasibility over a short sweep | Start from the regime schedule in Table 3; reduce $\epsilon_\kappa$ if infeasibility rises without reducing violation. |
| Envelope check | Evidence of strong state-dependent amplification or repeated QP relaxation | Treat the unit as outside the calibrated envelope and switch to conservative operation instead of increasing $\epsilon_\kappa$. |
| Online monitoring | Drift in residual coverage, intervention rate, or active enthalpy margin | Recalibrate the GP, repeat the local $\epsilon_\kappa$ sweep, or hold the previous safe schedule until coverage recovers. |

Unsafe margin values are evaluated only in offline replay or supervised commissioning windows, not during ordinary plant operation. Once the plant is in service, updates to the

GP or $\epsilon_\kappa$ schedule should be gated by residual-coverage checks, QP fallback thresholds, and operator-approved alarm logic.

**GP calibration strategy.** Three modes define a clear deployment envelope:

(i) **Scenario-specific GP** (recommended): calibrate on commissioning data from the target operating condition. This gives the tightest residual model and enables the lowest practical $\epsilon_\kappa$.

(ii) **Online GP adaptation**: update incrementally during operation. This enables deployment without prior scenario knowledge, at the cost of a transient period before the GP becomes accurate.

(iii) **Generic mixed GP**: train on pooled data from multiple scenarios without scenario labels. This mode is outside the deployment envelope because silent calibration failure produces 82.7–99.7% violation (Phase 4, 3rd-order).

$\epsilon_\kappa$ **selection.** Once the GP is calibrated, $\epsilon_\kappa$ should be set according to the expected uncertainty structure and strength (Sec. 4.4):

(i) **Additive uncertainty** ($|\delta|$ bounded, scenarios S1, S2, S4–S6): use $\epsilon_\kappa \in [0, 0.3]$. GP mean correction alone provides sufficient safety in the tested regime, so the margin can be set to zero for minimal intervention.

(ii) **State-dependent uncertainty in the $\epsilon_\kappa$-useful regime** ($0.5 \leq \gamma < 1.5$, scenario S3): use $\epsilon_\kappa \in [0.1, 0.3]$. A positive margin is needed because GP mean correction alone fails at $\gamma \geq 0.5$ (>8% violation). As coupling strengthens, the safe $\kappa$ range narrows; $\epsilon_\kappa$=1.0 over-constrains the QP in the tested cases.

(iii) **Strong state-dependent uncertainty beyond the envelope** ($\gamma \geq 1.5$, scenario S3): treat positive margins as counterproductive. The HOCBF Lie-derivative chain cannot compensate for the perturbation amplification rate, and any $\kappa$>0 produces 75–99% violation in this diagnostic. QP infeasibility should trigger fallback to a conservative operating regime rather than increasing $\kappa$.

(iv) **No uncertainty** (nominal, load following): set $\epsilon_\kappa = 0$ and omit the GP when pure HOCBF is sufficient.

Prior cross-scenario validation (Phase 4, 3rd-order) confirmed the out-of-distribution failure mode (S3→S1: 99.9% violation). The mechanism is explicit: $\sigma_{\mathrm{GP}}$ measures spatial distance to training data, not functional distance between perturbation structures. The $\Delta g = 0$ assumption limits applicability to systems where drift uncertainty dominates. QP infeasibility is a one-sided diagnostic: it detects overestimated $\sigma_{\mathrm{GP}}$ (overly conservative margin), but not underestimated $\sigma_{\mathrm{GP}}$ (unsafe failure). The practical recommendation is to calibrate the GP on scenario-representative data, select $\epsilon_\kappa$ from the regime schedule above, and bound the final setting by the coupling-strength envelope in Sec. 4.4.

Additional results (load-following AGC performance under $\Phi$-scaled dynamics, per-constraint violation decomposition across all methods and conditions, full $\epsilon_\kappa$ sweep data, PPO-Lagrangian and PPO-CBF baselines, and triple-integrator $m = 3$ validation) are provided in the Supplemental Material.

# 5 Conclusion

This paper presented RoCBF-Net as a tunable GP-HOCBF safety filter for a 1000 MW ultra-supercritical boiler-turbine measurement and control loop. The filter sits between an upstream controller and the actuators, using measured pressure, enthalpy, power, fuel, and delay states to project the proposed action into the process-safe set.

The main engineering conclusion is that robustness margins should be commissioned, not fixed. On the 5th-order nonlinear CCS benchmark with $\Phi$-scaled control effectiveness, GP mean correction alone reduces HOCBF violations by more than 500 times and achieves $\leq 0.18\%$ violation in six of seven uncertainty scenarios. For additive perturbations, the recommended setting is $\epsilon_\kappa = 0$ because GP mean correction provides the useful compensation and the margin mainly reduces QP feasibility. For moderate state-dependent coupling, $\epsilon_\kappa = 0.1$ restores zero violation where the GP-only filter fails. The full theoretical setting $\epsilon_\kappa = 1.0$ is not preferred in any tested condition because it can over-constrain the QP and trigger fallback. The coupling-strength sweep defines the deployment envelope: small positive margins are useful for $0.5 \leq \gamma < 1.5$, while beyond $\gamma \geq 1.5$ the margin should be disabled and infeasibility should be treated as a fallback or alarm signal.

The computational result supports real-time use in the simulated DCS loop. The JIT-compiled GP-HOCBF-QP pipeline runs at about 25 ms per step on an RTX 4090 GPU, within the 1 s sampling budget and approximately ten times faster than the implemented SLSQP-based NMPC benchmark. This timing comparison is against the implemented research NMPC baseline, not against optimized proprietary industrial MPC packages.

The validation boundary should guide use of the results. All evidence is from simulation benchmark experiments rather than plant trials, so the commissioning schedule is a pre-deployment validation protocol rather than a field authorization. The model-mismatch formulation assumes $\Delta g = 0$, so actuator-gain uncertainty and severe valve nonlinearities require additional treatment. The invariance result is continuous-time and probabilistic, while the implemented controller is sampled-data with zero-order hold; inter-sample behavior is checked empirically, but a formal discrete-time invariance theorem is not claimed. The next validation steps are hardware-in-the-loop testing, plant historian calibration, actuator-gain uncertainty, and certified sampled-data GP-HOCBF filtering.

# Statements and Declarations

## Ethical considerations

Not applicable. This study did not involve human participants, human data, human tissue, or animal experiments.

## Consent to participate

Not applicable.

## Consent for publication

Not applicable.

## Declaration of conflicting interests

The authors declare no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

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## Author contributions

Zhuang Shao: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Visualization, Writing – original draft, Writing – review & editing. Lijun Lei: Resources, Supervision, Project administration. Peng Wang: Formal analysis, Writing – review & editing. Liang Zheng: Resources, Supervision. Jie Zhou: Investigation, Data curation.

## Data availability

The source code, simulation scripts, trained GP configurations, and plotting scripts required to reproduce the results are available in the GitHub repository. No proprietary plant historian data, human data, or third-party operational datasets were used in this study.

## Declaration of generative AI and AI-assisted technologies

During the preparation of this work, the authors used ChatGPT for language polishing and manuscript formatting. The authors reviewed and edited all content and take full responsibility for the article.

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